
Algorithm 1 Knoll-Yliluoma positional dithering

Require: Target color $\mathbf{T} \in \mathbb{R}^3$

- 1: Palette $\{\mathbf{p}_j\}_{j=1}^N \subset \mathbb{R}^3$
- 2: Candidate count $C \in \mathbb{Z}_{>0}$
- 3: Pixel coordinates (x, y) and stride

▷ stride is typically the image width

Ensure: Output color $\mathbf{q} \in \mathbb{R}^3$

- 4: $\mathbf{e} \leftarrow \mathbf{0} \in \mathbb{R}^3$ ▷ Initialize error accumulator
 - 5: **for** $c = 1, \dots, C$ **do**
 - 6: $\mathbf{t}_c \leftarrow \text{clamp}(\mathbf{T} + \mathbf{e}, \mathbf{0}, \mathbf{1})$ ▷ Compute target with error feedback
 - 7: $k_c \leftarrow \arg\min_{1 \leq j \leq N} \|\mathbf{p}_j - \mathbf{t}_c\|^2$ ▷ Find nearest palette color
 - 8: $\mathbf{q}_c \leftarrow \mathbf{p}_{k_c}$ ▷ Store candidate color
 - 9: $\mathbf{e} \leftarrow \mathbf{T} - \mathbf{q}_c$ ▷ Update accumulated error
 - 10: $\ell_c = \text{luma}(\mathbf{q}_c)$, $\text{luma}(\mathbf{c}) = 0.299c_r + 0.587c_g + 0.114c_b$ ▷ Compute candidate luminances
 - 11: Sort candidates $\{\mathbf{q}_c\}_{c=1}^C$ by increasing luminance ℓ_c to obtain a deterministic order.
 - 12: $s \leftarrow ((x + y \cdot \text{stride}) \bmod C) + 1$ ▷ Select candidate index
 - 13: **return** $\mathbf{q}_s$
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Algorithm 2 4×4 ordered dithering (Bayer matrix)

Require: Target color $\mathbf{T} \in \mathbb{R}^3$

- 1: Palette $\{\mathbf{p}_j\}_{j=1}^N \subset \mathbb{R}^3$
- 2: Pixel coordinates (x, y)

Ensure: Output color $\mathbf{q} \in \mathbb{R}^3$

- 3: $\mathbf{B} = \frac{1}{16} \begin{bmatrix} 0 & 8 & 2 & 10 \\ 12 & 4 & 14 & 6 \\ 3 & 11 & 1 & 9 \\ 15 & 7 & 13 & 5 \end{bmatrix}$ ▷ 4×4 Bayer threshold matrix
 - 4: $\tau \leftarrow B_{x \bmod 4, y \bmod 4} - \frac{1}{2}$ ▷ Lookup centered threshold value
 - 5: $\mathbf{t} \leftarrow \text{clamp}(\mathbf{T} + \tau, \mathbf{0}, \mathbf{1})$ ▷ Apply ordered dither offset
 - 6: $k \leftarrow \arg\min_{1 \leq j \leq N} \|\mathbf{p}_j - \mathbf{t}\|^2$ ▷ Select nearest palette color
 - 7: **return** $\mathbf{q} \leftarrow \mathbf{p}_k$
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